

Notes: Sørensen (JF, 2007)
How Smart Is Smart Money? A Two-Sided Matching Model of Venture Capital

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1 Big picture

- **Question.** Do experienced venture capitalists improve company outcomes, or do they simply obtain access to better companies?
- **Outcome.** The main outcome is whether a portfolio company eventually goes public through an IPO.
- **Main empirical problem.** More experienced VCs invest in better companies, so a regression of IPO on VC experience confounds investor influence with sorting.
- **Core method.** The paper builds a structural two-sided matching model that uses the characteristics of other agents in the market to separate influence from sorting.
- **Main finding.** Both influence and sorting are significant, but sorting is almost twice as important as influence in explaining differences in IPO rates across investors.
- **Economic takeaway.** “Smart money” is smart both because experienced VCs add value and because they have better access to promising ventures.

2 Incremental contribution of the paper

- It separates *influence* from *sorting*, two mechanisms that are usually bundled together in venture capital performance studies.
- It introduces a structural matching correction for entrepreneurial finance, avoiding the need for a conventional external instrument for VC experience.
- It adapts the Gale–Shapley many-to-one matching framework to VC-company matching.
- It quantifies the relative importance of value added and access to better deals: sorting explains more of the observed IPO-rate gap.
- It shows how the characteristics of other market participants create identifying variation through the equilibrium matching structure.

3 Data and measurement

3.1 VentureXpert sample

The data come from SDC VentureXpert. The full database contains 22,747 VC investments from 1975 to 1995. The final estimation sample contains 1,666 young companies receiving first-round lead-investor VC financing between 1982 and 1995.

Interpretation: The sample isolates the initial lead investor-company match, which is the object modeled by the two-sided matching framework.

3.2 Markets and feasible matches

The baseline market is a half-year by state market, focusing on California and Massachusetts. Each market defines the set of investors and companies that could feasibly match.

Interpretation: The market definition matters because the structural likelihood uses the set of observed and unobserved feasible matches inside each market.

3.3 Outcome and key variables

The outcome is IPO_{ij} , an indicator equal to one if the company goes public. Investor experience is the number of previous investment rounds made by the VC firm. Controls include fund size, company stage, state, industry, and investment year.

Interpretation: Experience is both a potential causal input and an endogenous matching outcome.

4 Models and empirical specifications

4.1 Outcome equation

$$Y_{ij} = X_{ij}\beta + \varepsilon_{ij}, \quad IPO_{ij} = \mathbf{1}\{Y_{ij} \geq 0\}.$$

Interpretation: The coefficient on experience in the outcome equation is the VC influence effect after correcting for sorting.

4.2 Matching model

The set of investors is I , the set of companies is J , and potential matches are $M = I \times J$. Investor i has quota q_i . A matching is denoted $\mu \subset M$.

4.3 Valuation and preferences

$$\Pi_i(\mu(i)) = \lambda \sum_{j \in \mu(i)} V_{ij}, \quad \Pi_j = (1 - \lambda)V_{\mu(j),j}.$$

Interpretation: A common match valuation ranks matches for both sides. The fixed sharing rule rules out transfers and gives uniqueness.

4.4 Valuation equation

$$V_{ij} = W_{ij}\alpha + \eta_{ij}.$$

Interpretation: The valuation equation captures preferences over matches and therefore the sorting mechanism.

4.5 Stability restrictions

For unmatched pairs, stability requires $V_{ij} < \bar{V}_{ij}$. For matched pairs, stability requires $V_{ij} > \underline{V}_{ij}$. These inequalities define the likelihood of the observed matching.

Interpretation: Observing who matched with whom and who did not match is informative about latent preferences.

5 Identification and empirical design

The core problem is endogenous matching. Better companies may choose better VCs, and better VCs may have access to better companies. The structural model uses the equilibrium assignment itself to address this endogeneity.

- **Source of variation.** Other investors and companies in the same market change feasible stable matchings and relative ranks.
- **Instrument-like logic.** Other agents' characteristics shift the probability of matching with an experienced investor but do not directly determine the IPO outcome of a fixed potential match.
- **Observed sorting check.** Table IV shows that late-stage and biotech firms match with more experienced VCs.
- **Unobserved sorting check.** Figure 4 and the positive covariance between valuation and outcome errors show sorting on unobserved characteristics.
- **Robustness.** Table VII varies the market definition and preserves the main influence-versus-sorting conclusion.

Interpretation: The identification strategy is not a natural experiment. It is a structural selection correction using two-sided matching restrictions.

6 Tables and figures

6.1 Figure 1: Matching market with four companies and two or three investors

Read: The figure shows how a naive regression can falsely attribute better company quality to investor experience. Adding an investor changes stable matching and generates counterfactual information.

Economic message: Market composition is the identification idea: other agents shift assignments and help separate sorting from influence.

		Company			
		4 ($X_4=4$)	3 ($X_3=3$)	2 ($X_2=2$)	1 ($X_1=1$)
Investor	C ($X_C=3$)	-	-	-	-
	B ($X_B=2$)	40	30	-	-
	A ($X_A=1$)	-	-	20	10
		-	-	-	10

Figure 1. Matching market with four companies and two or three investors. Bold numbers indicate outcomes from a matching market with investors A and B. Remaining (not bold) numbers are outcomes from a matching market with investors A, B, and C.

Figure 1: Matching market with four companies and two or three investors. The bold numbers show observed matches in the two-investor market; remaining numbers show outcomes in the larger market.

6.2 Table I: Descriptive Statistics; Figure 2: Number of companies; Figure 3: IPO rate

Read: Table I reports 1,666 companies and an average IPO rate of 26.9%. Figures 2 and 3 show that companies backed by more experienced VCs are fewer but have higher IPO rates.

Economic message: The raw experience-performance pattern motivates the decomposition but cannot be interpreted causally.

Table I
Descriptive Statistics

Reported figures are averages over the observed investments in the sample. The sample contains investments in 1,666 companies. The variable *IPO* equals one if the investment results in a public offering and zero otherwise. *YEAR* contains the year of the investment. *FUND_SIZE* is the amount of capital available to the investor, measured as the total amount of investments made by the fund. *STATE* equals zero if the company is located in Massachusetts and equals one if the company is located in California. *STAGE* equals zero if the company is an early-stage company and equals one if it is a late-stage company. *I.XXXX* are six industry controls. *TOTAL_EXPERIENCE* is the experience of the investor, measured as the number of previous investments made by the VC firm.

	OBS	MEAN	STD. DEV.	MIN	MAX
<i>IPO</i>	1,666	0.269	0.444	0	1
<i>YEAR</i>	1,666	1,988.340	4.219	1982	1995
<i>FUND_SIZE</i> (\$MIL)	1,666	130.556	210.650	1.4	2,113.9
<i>STATE</i>	1,666	0.791	0.414	0	1
<i>STAGE</i>	1,666	0.175	0.382	0	1
<i>I.COMMUNICATION</i>	1,666	0.131	0.338	0	1
<i>I.COMPUTER</i>	1,666	0.421	0.494	0	1
<i>I.ELECTRONICS</i>	1,666	0.101	0.301	0	1
<i>I.BIOTECHNOLOGY</i>	1,666	0.062	0.241	0	1
<i>I.MEDICAL</i>	1,666	0.134	0.341	0	1
<i>I.OTHER</i>	1,666	0.150	0.357	0	1
<i>TOTAL_EXPERIENCE</i>	1,666	69.562	66.178	0	443

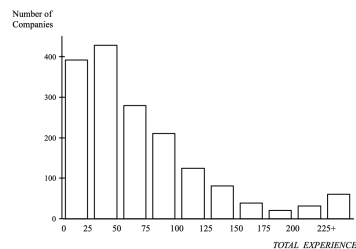


Figure 2. Number of companies in each group. The 1,666 companies are grouped into 10 groups according to the experience of the investor. The figure illustrates the number of companies in each group.

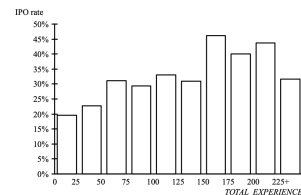


Figure 3. IPO rate in each group. The 1,666 companies are grouped into 10 groups according to the experience of the investor. The figure shows IPO rates of companies in each group.

(a) Table I: Descriptive Statistics. Sample size, IPO rate, experience, and controls.

(b) Figure 2: Number of companies. Distribution across VC experience bins.

(c) Figure 3: IPO rate. IPO rates by VC experience bins.

6.3 Table II: Probit Estimates; Table III: Probit Estimates with Alternative Specifications

Read: The Probit tables show a positive relationship between VC experience and IPO probability even after controls.

Economic message: These are total effects combining sorting and influence.

Table II: Probit Estimates

The table contains estimated coefficients from three specifications of a Probit model. The dependent variable is *IPO*. The dummy variables *I.XXXX* control for individual years, with omitted base year 1982. The reported coefficients are ML estimates, and *dF/dX* is the marginal effect of a change in the variable evaluated at the sample average. The marginal effects for binary variables (marked *) are the effects of a discrete change. *STD. ERR.* and *P-VAL.* are standard errors and statistical significance of marginal effect. The first three columns with standard errors and *p*-values are evaluated with clustering at the investor level. The last column with *p*-values is evaluated without clustering. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Specification 1			Specification 2			Specification 3		
	COEF	STD. ERR.	P-VAL.	COEF	STD. ERR.	P-VAL.	COEF	STD. ERR.	P-VAL.
<i>TOTAL_EXPERIENCE</i>	0.000657	(0.000250)	0.01***	0.048717	(0.015289)	0.00***	0.000657	(0.000248)	0.01***
<i>LOG_EXPERIENCE (-1)</i>									
<i>FUND_SIZE</i> (\$ MIL)	0.000139	(0.00071)	0.05**	0.000128	(0.00066)	0.04**	0.000141	(0.00069)	0.04**
<i>STAGE</i>	0.0020	(0.0030)	0.04*	0.0071	(0.0352)	0.04**	0.0079	(0.0309)	0.05*
<i>I.COMMUNICATION</i>	0.2741	(0.0480)	0.00***	0.2775	(0.0477)	0.00***	0.2692	(0.0486)	0.00***
<i>I.COMPUTER</i>	0.1487	(0.0428)	0.00***	0.1514	(0.0428)	0.00***	0.1418	(0.0428)	0.00***
<i>I.ELECTRONICS</i>	0.2189	(0.0651)	0.00***	0.2203	(0.0652)	0.00***	0.2052	(0.0643)	0.00***
<i>I.BIOTECH</i>	0.4359	(0.054)	0.00***	0.4334	(0.054)	0.00***	0.4328	(0.0572)	0.00***
<i>I.MEDICAL</i>	0.1943	(0.0472)	0.00***	0.1973	(0.0472)	0.00***	0.1949	(0.0490)	0.00***
<i>STATE</i>	0.000614	(0.02180)	1.90	-0.000514	(0.021194)	0.98	0.001116	(0.02470)	0.92
<i>YEAR</i>									
<i>Y.1983</i>	-0.0400	(0.0306)	0.23	-0.0516	(0.0390)	0.21	-0.0443	(0.0353)	0.20
<i>Y.1984</i>	-0.0400	(0.0442)	0.38	-0.0534	(0.0435)	0.24			
<i>Y.1985</i>	-0.1167	(0.0440)	0.03**	-0.1265	(0.0432)	0.02*			
<i>Y.1986</i>	0.0559	(0.0630)	0.66	0.0129	(0.0623)	0.84			
<i>Y.1987</i>	-0.0005	(0.0562)	0.91	-0.0202	(0.0549)	0.72			
<i>Y.1988</i>	0.0239	(0.054)	0.65	0.0081	(0.0540)	0.88			
<i>Y.1989</i>	-0.0644	(0.0487)	0.22	-0.0769	(0.0470)	0.13			
<i>Y.1990</i>	-0.0031	(0.0568)	0.19	-0.0071	(0.0551)	0.12			
<i>Y.1991</i>	0.0731	(0.0751)	0.31	0.0547	(0.0764)	0.46			
<i>Y.1992</i>	-0.0072	(0.0496)	0.84**	-0.1997	(0.0393)	0.02**			
<i>Y.1993</i>	0.0298	(0.0631)	0.64	0.0090	(0.0610)	0.88			
<i>Y.1994</i>	-0.1963	(0.0497)	0.05**	-0.1209	(0.0425)	0.02**			

(a) Table II: Probit Estimates. Reduced-form IPO regressions.

(b) Table III: Alternative Probit Estimates. Experience remains positive with controls and alternative forms.

6.4 Table IV: OLS Estimates

Read: Late-stage and biotech companies match with more experienced VCs.

Economic message: This confirms sorting on observed characteristics.

Table IV
OLS Estimates

The table presents coefficients from three OLS regressions. The dependent variable is *TOTAL_EXPERIENCE_AVG_EXPERIENCE* contains average experience of the investors in each market. *STATE_CONTROL* controls for the location of the company. *YEAR_CONTROLS* controls for individual years. STD. ERR. is the robust standard error of the coefficients. *, **, and*** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Specification 1		Specification 2		Specification 3	
	COEF.	STD. ERR.	COEF.	STD. ERR.	COEF.	STD. ERR.
<i>YEAR</i>	5.084	(0.364)***	-0.114	(0.640)		
<i>STAGE</i>	19.153	(4.025)***	18.552	(3.916)***	18.653	(3.943)***
<i>I.COMMUNICATION</i>	3.068	(5.787)	2.162	(5.631)	2.132	(5.680)
<i>I.COMPUTER</i>	-1.246	(4.587)	-1.843	(4.463)	-1.924	(4.532)
<i>I.ELECTRONICS</i>	5.170	(6.221)	3.436	(6.054)	3.435	(6.108)
<i>I.BIOTECHNOLOGY</i>	11.960	(7.351)	14.321	(7.156)**	14.520	(7.242)**
<i>I.MEDICAL</i>	5.765	(5.738)	4.404	(5.584)	4.498	(5.651)
<i>AVG_EXPERIENCE</i>			1.000	(0.103)***	0.997	(0.116)***
<i>CONSTANT</i>	-10044.3	(722.8)***	222.4	(1267.5)	-4.5	(13.9)
<i>STATE_CONTROL</i>	no		no		yes	
<i>YEAR_CONTROLS</i>	no		no		yes	
R^2	0.12		0.17		0.17	
N	1666		1666		1666	

Table IV: OLS Estimates. Investor experience regressed on company characteristics.

6.5 Table V: Bayesian Estimates of Structural Model; Figure 4: Error Terms of Matched Pairs

Read: Table V estimates valuation and outcome equations jointly. Figure 4 shows a positive relation between investor relative experience and valuation residuals.

Economic message: Experience adds value, but unobserved high-quality matches also sort to experienced VCs.

Table V
Bayesian Estimates of Structural Model

The table presents Bayesian estimates of the parameters in the two equations in the structural model. The dependent variable in the outcome equation is IPO, and the dependent variable in the valuation equation is the latent valuation variable. A detailed description of the construction of all variables is in the main text. MEAN, MEDIAN, and STD. DEV. are, respectively, the mean, median, and standard deviation of the simulated posterior distributions of the parameters. The reported coefficients for the outcome equation are normalized to be comparable to standard Probit estimates. dF/dX is the marginal effect. The marginal effects are evaluated at the mean of the posterior. The marginal effects for variables marked $\circ$ are effects of a discrete change. dP/dW is the marginal change in the probability of preferring one of two otherwise identical matches with a change in the variable (the precise definition is in the text). Estimates are based on 1,200,000 simulations of the posterior distribution. The initial 400,000 simulations are discarded for burn-in. *, **, and *** denote that zero is not contained in the 10%, 5%, and 1% credible interval, respectively.

	MEAN	MEDIAN	dF/dX	dP/dW	STD. DEV.
Outcome equation					
<i>TOTAL EXPERIENCE</i>	0.001385	0.001385	0.000346		(0.000302)***
<i>FUND_SIZE (\$ MIL)</i>	0.000328	0.000328	0.000082		(0.000092)***
<i>YEAR</i>	-0.000895	-0.000907	-0.000223		(0.000527)*
<i>STAGE</i> $\circ$	0.1843	0.1849	0.0486		(0.0499)***
<i>I.COMMUNICATION</i> $\circ$	0.7101	0.7076	0.2183		(0.0785)***
<i>I.COMPUTER</i> $\circ$	0.4092	0.4082	0.1052		(0.0676)***
<i>I.ELECTRONICS</i> $\circ$	0.5612	0.5603	0.1687		(0.0838)***
<i>I.BIOTECHNOLOGY</i> $\circ$	1.1274	1.1269	0.3859		(0.0890)***
<i>I.MEDICAL</i> $\circ$	0.5238	0.5226	0.1538		(0.0793)***
<i>CONSTANT</i>	0.1763	0.1931			(1.0458)
Valuation Equation					
<i>TOTAL EXPERIENCE</i>	0.0290	0.0281		0.0082	(0.0127)***
<i>FUND_SIZE (\$ MIL)</i>	0.0465	0.0418		0.0131	(0.0251)***
<i>STAGE</i> $\circ$	0.1657	0.1644		0.0466	(0.0330)***
<i>I.COMMUNICATION</i> $\circ$	0.0252	0.0259		0.0071	(0.0434)
<i>I.COMPUTER</i> $\circ$	0.0074	0.0082		0.0021	(0.0360)
<i>I.ELECTRONICS</i> $\circ$	0.0272	0.0276		0.0077	(0.0462)
<i>I.BIOTECHNOLOGY</i> $\circ$	0.2808	0.2801		0.0787	(0.0567)***
<i>I.MEDICAL</i> $\circ$	0.0713	0.0718		0.0201	(0.0446)
Variance					
<i>COVARIANCE</i>	0.2222	0.2208			(0.0670)***

(a) **Table V: Structural Model.** Outcome, valuation, and covariance estimates.

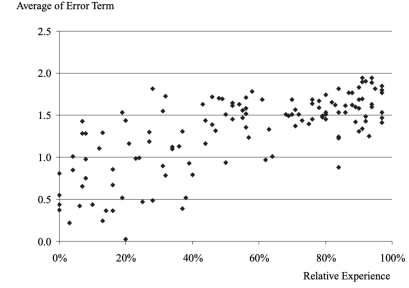


Figure 4. Error terms of matched pairs. For each investor in each market, the average of the error terms in the valuation equation is plotted as a function of the investor's experience relative to the other investors in the same market. Relative experience is the investor's experience as a percentile of the experience of all the investors in the market. For the most experienced investor in the market, the relative experience is one. The average is calculated as the average of the error terms in the matches that are made by the investor. Only investors who participate in two or more matches in the market are included in the figure.

(b) **Figure 4: Error Terms.** Sorting on unobserved valuation residuals.

6.6 Table VI: Alternative Specifications; Figure 5: Decomposition of Influence and Sorting

Read: Table VI checks alternative outcome specifications. Figure 5 decomposes raw IPO differences into influence and sorting.

Economic message: Sorting explains more of the raw IPO gap than direct investor influence.

Table VI
Bayesian Estimates of Outcome Equation under Alternative Specifications

The table contains Bayesian estimates of the coefficients in the outcome equations for three specifications of the structural model. The dependent variable in the outcome equation is IPO. INV1, INV2, up to INV5 are investor-specific controls for the five most experienced investors in the data. A detailed description of the construction of all variables is in the main text. MEAN and STD. DEV. are, respectively, the mean and standard deviation of the simulated posterior distributions of the coefficients. The reported coefficients for the outcome equation are normalized to be comparable to standard Probit estimates. dF0X is the marginal effect, evaluated at the MEAN of the posterior. The marginal effect for discrete variables marked * is the effect of a discrete change. Estimates are based on 1,200,000 simulations of the posterior distribution. The initial 400,000 simulations are discarded for burn-in. *, **, and *** denote that zero is not contained in the 10%, 5%, and 1% credible interval, respectively.

	Specification 1			Specification 2			Specification 3		
	MEAN	dF0X	STD. DEV.	MEAN	dF0X	STD. DEV.	MEAN	dF0X	STD. DEV.
Outcome equation									
TOTAL EXPERIENCE	0.001808	0.000441	(0.000483)***	0.001385	0.000346	(0.000302)***	0.000260	0.000068	(0.000435)
FUND SIZE (\$ MIL)				0.000328	0.000082	(0.000092)***	0.000732	0.000196	(0.000179)***
YEAR				-0.000895	-0.000223	(0.000377)*	-0.000884	-0.000230	(0.000751)
STAGE				0.1843	0.0486	(0.0489)***	0.1747	0.0478	(0.0503)***
I.COMMUNICATION				0.7101	0.2183	(0.0785)***	0.6897	0.2176	(0.0785)***
I.COMPUTER				0.4092	0.1052	(0.0676)***	0.3761	0.1005	(0.0680)***
I.ELECTRONICS				0.5612	0.1687	(0.0838)***	0.5370	0.1657	(0.0847)***
I.BIOTECHNOLOGY				1.1274	0.3859	(0.0880)***	1.1319	0.3949	(0.0912)***
I.MEDICAL				0.5238	0.1538	(0.0793)***	0.5120	0.1550	(0.0783)***
INV1							0.5415	0.1716	(0.1287)***
INV2							-0.0869	-0.0218	(0.1039)
INV3							0.3094	0.0906	(0.0818)***
INV4							0.1618	0.0451	(0.1013)
INV5							-0.7920	-0.1366	(0.3041)***
CONSTANT	-1.1180		(0.1013)***	0.1763		(1.0458)	0.2212		(1.4848)
Variance									
COVARIANCE	0.2499		(0.0732)***	0.2222		(0.0670)***	0.1925		(0.0651)***

(a) **Table VI: Alternative Specifications.** Structural outcome robustness.

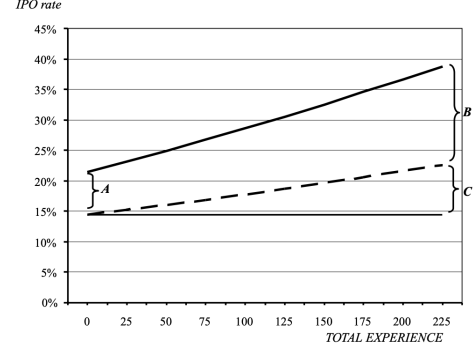


Figure 5: Decomposition of influence and sorting. The figure shows the IPO rates from the Probit model (Specification 1 in Table III) and the outcome equation of the structural model (Table V). The solid line indicates the IPO rate predicted by the Probit model. This rate is the empirical rate of the IPOs observed in the sample. The broken line represents the IPO rate from the outcome equation of the structural model. This is the IPO rate after controlling for the selection of the investments, and it represents the rate that would be observed if an average company were randomly assigned to investors with different degrees of experience.

(b) **Figure 5: Decomposition.** Solid line is total Probit effect; dashed line is influence after sorting correction.

6.7 Table VII: Bayesian Estimates under Alternative Market Specifications

Read: The structural model is re-estimated under alternative definitions of markets.

Economic message: The central conclusion is robust to the exact market definition.

Table VII
Bayesian Estimates under Alternative Market Specifications

The table contains Bayesian estimates of the outcome equation of the structural model. The model is estimated with four market specifications. The first specification takes a market to be delineated by a half-year and a state (either CA or MA). The second specification expands the time dimension, and takes a market to be delineated by a whole year and a state. The third specification expands the geographical dimension and takes a market to be delineated by a half-year, but treats CA and MA as one market. The final specification takes a market to be delineated by a whole year and treats CA and MA as one market. The dependent variable is IPO. A description of the variables is presented in Table I and in the main text. MEAN and STD. DEV. are, respectively, the mean and standard deviation of the simulated posterior distributions. The reported coefficients for the outcome equation are normalized to be comparable to standard Probit estimates. Estimates are based on 1,200,000 simulations of posterior distribution. The initial 400,000 simulations are discarded for burn-in. *, **, and *** denote that zero is not contained in the 10%, 5%, and 1% credible interval, respectively.

	Normal Mkt Def		Expanded Years		Expanded States		Expanded Years and States	
	MEAN	STD. DEV.	MEAN	STD. DEV.	MEAN	STD. DEV.	MEAN	STD. DEV.
Outcome equation								
TOTAL EXPERIENCE	0.001385	(0.000302)***	0.001093	(0.000310)***	0.001079	(0.000310)***	0.001322	(0.000300)***
FUND SIZE (\$ MIL)	0.000328	(0.000092)***	0.000316	(0.000080)***	0.000322	(0.000080)***	0.000370	(0.000098)***
YEAR	-0.000895	(0.000527)*	-0.000924	(0.000895)	-0.001410	(0.000751)*	-0.001089	(0.000506)***
STAGE	0.184	(0.050)***	0.186	(0.050)***	0.187	(0.049)***	0.175	(0.049)***
I.COMMUNICATION	0.710	(0.079)***	0.712	(0.077)***	0.711	(0.078)***	0.706	(0.078)***
I.COMPUTER	0.409	(0.068)***	0.413	(0.067)***	0.413	(0.068)***	0.415	(0.067)***
I.ELECTRONICS	0.561	(0.084)***	0.572	(0.083)***	0.570	(0.083)***	0.565	(0.083)***
I.BIOTECHNOLOGY	1.127	(0.089)***	1.141	(0.091)***	1.143	(0.090)***	1.118	(0.089)***
I.MEDICAL	0.524	(0.079)***	0.536	(0.079)***	0.536	(0.078)***	0.525	(0.078)***
CONSTANT	0.176	(1.046)	0.065	(1.783)	1.039	(1.500)	0.319	(1.024)
Variance								
COVARIANCE	0.222	(0.067)***	0.316	(0.084)***	0.311	(0.084)***	0.323	(0.091)***
N	1666		1666		1666		1666	
Markets	56		28		28		14	

Table VII: Alternative Market Specifications. Robustness to broader time and state definitions.

7 Short summary

- Experienced VCs have higher IPO rates.
- The raw relationship combines influence and sorting.
- The paper models VC-company assignment as a stable two-sided matching problem.
- Observed matches and non-matches identify preferences over matches.
- Structural estimates show that experience adds value, but sorting is almost twice as important as influence.
- Probit regressions overstate influence by up to about 90% if sorting is ignored.

8 Conclusion

8.1 Core conclusion

Both influence and sorting matter. Experienced VCs improve outcomes, but they also invest in better companies.

8.2 Economic intuition

Good companies prefer good VCs, and good VCs have stronger deal flow. The observed VC-company pair is therefore an equilibrium assignment, not an exogenous treatment.

8.3 Contribution to entrepreneurial finance

The paper clarifies that VC reputation and experience reflect both value added and access to better companies.

8.4 Limitations

The model is static, assumes complete information, uses IPO as the main outcome, and depends on market definitions.

8.5 Takeaway

In any setting with two-sided matching, partner quality is endogenous. Regressions of outcomes on partner characteristics should be interpreted as total assignment effects unless sorting is modeled.